COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	7/7/26	**DATE SUBMITTED:**	7/9/26
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Seatwork 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

1.

```cpp
#include <iostream>

int main() {
    int grades[3];
    int total = 0;
    float average;

    for (int i = 0; i < 3; i++) {
        std::cout << "Input quiz " << i + 1 << " score: ";
        std::cin >> grades[i];
    }

    for (int i = 0; i < 3; i++) {
        total += grades[i];
    }

    average = static_cast<float>(total) / 3;


    std::cout << "\nTotal of 3 quizzes: " << total << std::endl;
    std::cout << "Average quiz score: " << average << std::endl;

    return 0;
}
```

```
C:\Users\TIPQC\Documents\S
Input quiz 1 score: 90
Input quiz 2 score: 80
Input quiz 3 score: 70

Total of 3 quizzes: 240
Average quiz score: 80

--------------------------------
Process exited after 5.881 seconds with return value 0
Press any key to continue . . .
```

2.

```cpp
#include <iostream>

int main() {
    float prices[3] = {};
    float amount = 0, payment = 0, change = 0;
    int counter = 0;

    for (int i = 0; i != 3; i++) {
        std::cout << "Input item price: ";
        std::cin >> prices[i];
    }

    for (int j = 0; j != 3; j++) {
        amount += prices[j];

        std::cout << "\nItem " << j + 1 << ": " << prices[j] << std::endl;

        if (j == 2) {
            std::cout << "\nThe amount of the price is: " << amount;

            while (counter == 0) {
                std::cout << "\nPlease pay " << amount << ": ";
                float money;
                std::cin >> money;
                payment += money;

                if (payment < amount) {
                    std::cout << "\nInsufficient amount paid. Please add to your existing payment.";
                } else {
                    change = payment - amount;
                    std::cout << "\nYour change is: " << change;
                    counter = 1;
                }
            }
        }
    }

    return 0;
```

```
C:\Users\TIPQC\Documents\S
Input item price: 178
Input item price: 542
Input item price: 200

Item 1: 178

Item 2: 542

Item 3: 200

The amount of the price is: 920
Please pay 920: 500

Insufficient amount paid. Please add to your existing payme
nt.
Please pay 920: 420

Your change is: 0
--------------------------------
Process exited after 21.3 seconds with return value 0
Press any key to continue . . .
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Perform the following:

Create an Algorithm, Pseudocode, Flowchart, and C++ Program for the following:

1. Compute the total and average of 3 quizzes then display the total and average.

Algorithm:

1. Start
2. Declare an array named grades, integer variables total, and a float variable average.
3. Input the scores for the 3 quizzes and store them in the grades array.
4. Set the total to 0..
5. Add all three quiz scores to total.
6. Compute the average using the formula:
7. Display the total of the 3 quizzes.
8. Display the average quiz score.
9. Stop

Pseudocode:

```
START
        INITIALIZE grades[3], total = 0, average

        FOR i = 0 TO 2
                DISPLAY "Input quiz " + (i + 1) + " score: "
                INPUT grades[i]
        END FOR

        FOR i = 0 TO 2
                total = total + grades[i]
        END FOR

       average = total / 3
       DISPLAY "Total of 3 quizzes: " + total
       DISPLAY "Average quiz score: " + average

STOP
```

Flowchart:

2. Cashiering system that computes the total price of the 3 items bought, accepts payment and computes change. Display the transactions made.

Algorithm:

1. Start
2. Declare an array named prices and initialize all elements to zero.
3. Declare variables named amount, payment , change, and counter with values that is 0.
4. Input the prices of the 3 items and store them in the prices array.
5. Compute the total amount by adding all item prices.
6. Display each item and its price.
7. Display the total amount to be paid.
8. While the payment is less than the total amount.
9. Ask the user to enter a payment.

10. Add the entered payment to the current payment.
11. If the payment is still less than the total amount, display an Insufficient amount paid message.
12. Computes the change
13. Displays the change.
14. Stop

Pseudocode:

```
START
        INITIALIZE/DECLARE prices[3] = {0}
        INITIALIZE/DECLARE amount = 0, payment = 0, change = 0
        INITIALIZE/DECLARE counter = 0

        FOR i = 0 TO 2
                DISPLAY "Input item price: "
                INPUT prices[i]
        END FOR

        FOR j = 0 TO 2
                amount = amount + prices[j]
                DISPLAY "Item " + (j + 1) + ": " + prices[j]
        END FOR

        DISPLAY "The amount of the price is: " + amount

        WHILE counter == 0
                DISPLAY "Please pay " + amount + ": "
                INPUT money
                payment = payment + money
                IF payment < amount THEN
                        DISPLAY "Insufficient amount paid. Please add to your existing payment."
                END IF

                ELSE
                        change = payment - amount
                        DISPLAY "Your change is: " + change
                        counter = 1
                END ELSE
        END WHILE
STOP
```

Flowchart:

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Chatgpt, Perplexity.

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Code Assistance

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: Used to only locate the lines of errors for both AI.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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